Direction Matters: On the Implicit Regularization Effect of Stochastic Gradient Descent with Moderate Learning Rate

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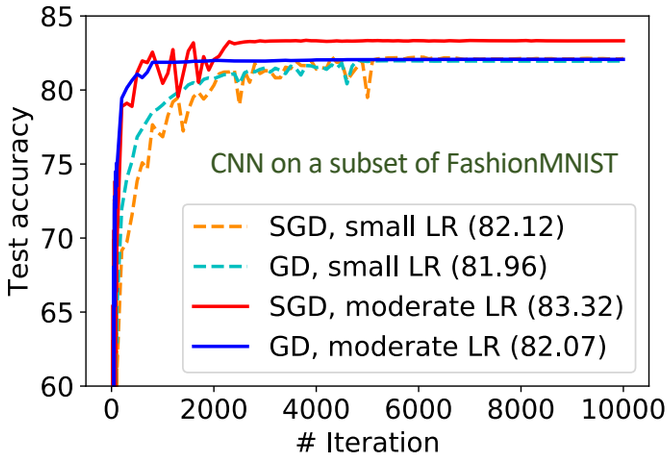


	Small LR	Moderate LR
GD	☹	☹
SGD	☹	☺

Additional notations

- $\lambda_i = \|x_i\|_2^2$, $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$
- P is the projection onto the column space of X
- Let γ_1 (γ_n) be the largest (smallest non-zero) eigenvalue of $XX^\top$, and v_1 (v_n) be the corresponding eigenvector

Setups

- Test data $x = \zeta \cdot \xi \in \mathbb{R}^d$, where $\zeta \in (0,1]$, $\xi \sim \mathcal{U}(S^{d-1})$.
- $\ell(x; w) = (w - w_*)^\top x x^\top (w - w_*)$
- Training data $\mathrm{X} = (x_1, \ldots, x_n)$, $i.i.d.$, $d \gg n$

Theorem 1: Consider SGD with batch size b and moderate LR,

$$\eta_t = \begin{cases} \eta \in \left({}^{b}/_{\lambda_1} + o(1), {}^{b}/_{\lambda_2} - o(1) \right), & t = 1, \ldots, T_1 \\ o(1), & t = T_1 + 1, \ldots, T_2 \end{cases}$$

then

$$\frac{P\left(w^{sgd} - w_*\right)}{\|P(w^{sgd} - w_*)\|_2} \to v_1 \pm o(1)$$

Theorem 2: Consider GD with moderate or small LR,

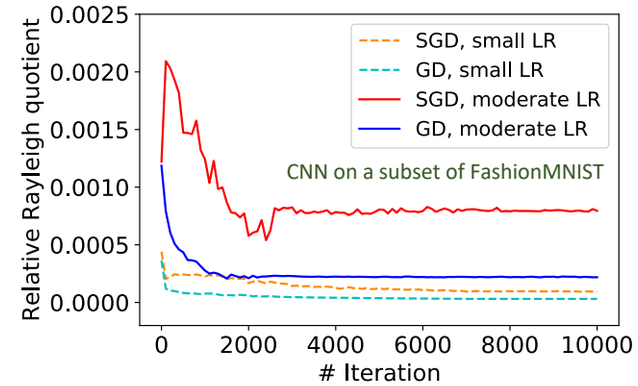
$$\eta_t \in \left(0, {}^{n}/_{2\lambda_2} - o(1)\right), \qquad t = 1, \ldots, T_2$$

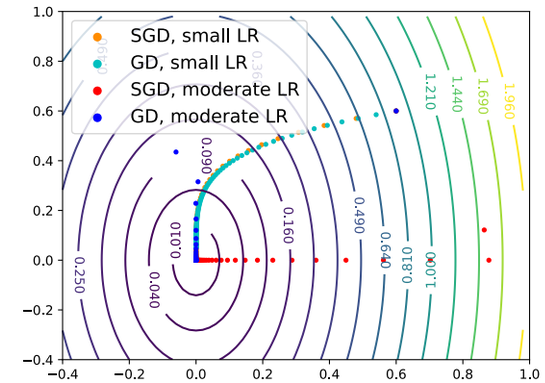
then

$$\frac{P\left(w^{gd} - w_*\right)}{\|P(w^{gd} - w_*)\|_2} \to v_n \pm o(1)$$

Theorem 3: Let $\Delta(w)$ be the estimation error and Δ_α^* be the optimal estimation error within an α-level set where the algorithms stop.

- For SGD with moderate LR, $\Delta\left(w^{sgd}\right) < \left(1 + o(1)\right) \cdot \Delta_\alpha^*$
- For GD with moderate/small LR, $\Delta\left(w^{gd}\right) > \left({}^{\gamma_1}/_{\gamma_n} - o(1)\right) \cdot \Delta_\alpha^*$





Take home

1. SGD + moderate LR: converges along large eigenvalue directions
2. GD + moderate/small LR: converge along small eigenvalue directions
3. The former directional bias benefits generalization